

Stephanie Wang

Education

Ph.D. and M.S. in Mathematics, UCLA, Eugene V. Cota-Robles Fellow. **2014-2020**
B.S. in Mathematics, National Taiwan University, *magna cum laude*. **2009-2013**

Experience

Research

Research Engineer – reference: Dr. Qingnan Zhou, Adobe Inc., Seattle, WA. **2024-present**
Geometry and physics research and software development, with a focus on geometry processing in machine learning and 3D modeling.

Postdoc – with Prof. Albert Chern, UCSD, San Diego, CA. **2020-2023**
Geometric deep learning, geometry optimization and physics-based simulation of various materials (solids, fluids, waves, cloth, hair, frictional contact, plasticity, fracture, *etc.*). Developing using Houdini and Python. Mentored students: [Mohammad Sina Nabizadeh](#), [Shiyang Jia](#), [Chad McKell](#), [Hang Yin](#), [Baichuan Wu](#).

Ph.D. Study – with Prof. Wilfrid Gangbo, UCLA, Los Angeles, CA. **2019-2020**
Regularity theory for minimizers of polyconvex functionals related to Navier-Stokes equation.

Exchange Study – with Prof. Johan Gaume, EPFL, Lausanne, Switzerland. **2019 summer**
Simulating snow and tire interaction using physics-based simulation, in collaboration with Michelin tires.

Ph.D Study – with Prof. Joseph Teran, UCLA, Los Angeles, CA. **2016-2019**
Physics-based simulation of various materials (fracture, cloth, hair, deformable body, *etc.*) using Material Point Method and Finite Element Method. Developing and maintaining large C++ library with high-performance numerical methods and parallel computing.

Tech Intern, Walt Disney Animation Studio, Burbank, CA. **2018 summer**
R&D for pioneering simulation technology in animated feature films, teaming with VFX artists and developing in C++ and Houdini HDK.

Teaching

Assistant Adjunct Professor / Instructor, UCLA Math Dept, Los Angeles, CA. **2019-2020**
Taught three courses: linear algebra, machine learning (remote) and multivariable calculus (remote).

Teaching Assistant, UCLA Math Dept, Los Angeles, CA. **2015-2020**
TA-ed courses: linear algebra, numerical methods (intro, intermediate, and advanced), C++ programming (intro, intermediate, and advanced).

Skills

Programming: C++ (Eigen, tbb, gdb, valgrind), Python (PyTorch, SciPy), MATLAB (CVX), $\LaTeX$, Vim, git, zsh, Houdini

Math: Optimization, differential geometry, solid and fluid dynamics, scientific computing

Languages: English and Mandarin Chinese (bilingual)

Technical communication: 11 papers in top journals and 18 talks at top conferences / institutes.

Selected Publications

[Implicit UVs: Real-time Semi-global Parameterization of Implicit Surfaces](#)

Baptiste Genest, Pierre Gueth, Jérémy Levallois, [Stephanie Wang](#)
Computer Graphics Forum (Eurographics 2025)

[Exterior Calculus in Graphics: Course Notes for a SIGGRAPH 2023 Course](#). [Stephanie Wang](#), Mohammad Sina Nabizadeh, Albert Chern. SIGGRAPH 2023.

[DeepCurrents: Learning implicit representations of shapes with boundaries](#). David Palmer, Dmitriy Smirnov, Stephanie Wang, Albert Chern, and Justin Solomon. CVPR 2022.

[Computing minimal surfaces with differential forms](#). [Stephanie Wang](#) and Albert Chern. ACM ToG (SIGGRAPH 2021).

[Simulation and visualization of ductile fracture with the material point method](#). [Stephanie Wang](#), Mengyuan Ding, Theodore F. Gast, Leyi Zhu, Steven Gagniere, Chenfanfu Jiang, and Joseph M. Teran. PACM-CGIT (SCA 2019 **Best Paper Award**).

...and 7 more papers published in top journals in computer graphics and other scientific fields.

Last updated: May 12, 2025.